CoverMap

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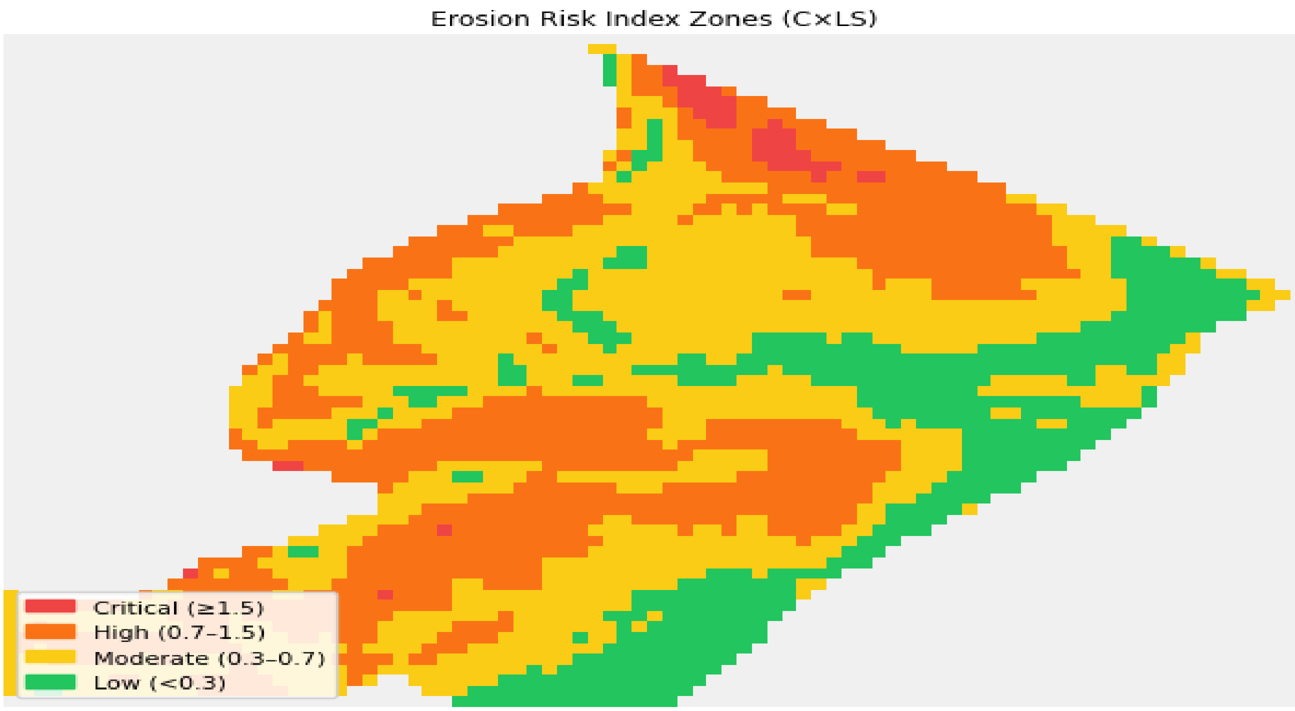
Field Cover Crop Assessment

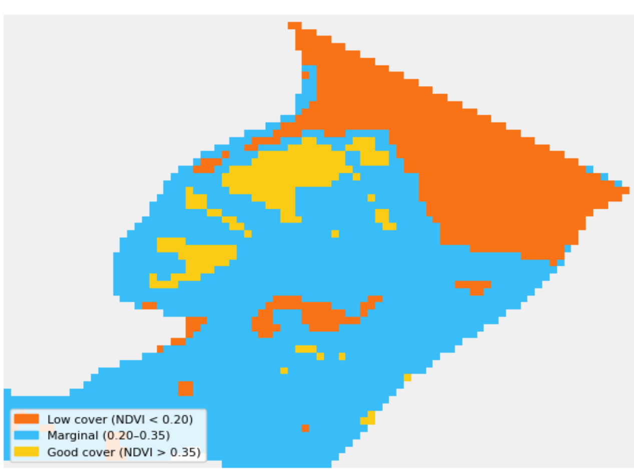
Field: Bin Field (2)	**Farm:** Bizy Farms Inc.	**County:** Shelby County, IA	**Report Date:** June 19, 2026
Previous crop: Corn	**Termination date:** May 1st.	**Dominant soil series:** Monona — K-factor: .32	**Previous crop / tillage:** Tillage — > 30% residue (conservation tillage)

NDVI: Apr 05, 2026 – Apr 20, 2026 | DEM: Iowa DNR 3m

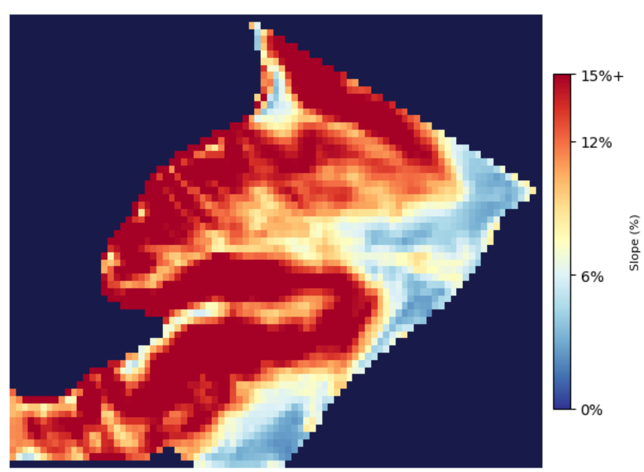
Field Risk Maps



Erosion Risk Index Zones (C×LS) — pixel-level RUSLE risk classification



NDVI Cover Quality — Low (<0.20) / Marginal / Good (>0.35)



Terrain Slope (% gradient) — Flat / Moderate / Steep

CoverMap Advisory & Recommendation

Erosion Concern: Moderate

Cover crop stand is variable across slope positions based on NDVI analysis. Erosion risk increases on steeper units during spring rainfall events.

Cover Crop Stand — NDVI Zone Summary

Zone	Acres	% of Field
Low Cover (NDVI < 0.20)	12.9	27%
Marginal (NDVI 0.20–0.35)	31.9	66%
Good Cover (NDVI > 0.35)	3.6	7%
Total	**48.3**	**100%**

Erosion Risk Zone Summary (C×LS)

Zone	C×LS Range	Acres	% of Field
Critical Risk	> 1.5	0.7	2%
High Risk	0.7–1.5	17.2	36%
Moderate Risk	0.3–0.7	21.0	43%
Low Risk	< 0.3	9.4	19%

Risk Index = C-factor (NDVI) × LS-factor (slope). Boundary-masked field pixels only. Critical >1.5 · High 0.7–1.5 · Moderate 0.3–0.7 · Low <0.3

NDVI imagery dated 6 scenes composited: Apr 08, 2026 – Apr 18, 2026. Field conditions may have changed since image capture. This report documents satellite-observed conditions only.

Cover Crop Stand Assessment — Satellite Documentation

Requirement	Data Source	Status
Cover crop present	Sentinel-2 NDVI > 0.20	■ NDVI 0.251 — cover crop confirmed
Image date	GEE metadata	6 scenes composited: Apr 08, 2026 – Apr 18, 2026
Estimated biomass	NDVI proxy	~700–1650 lb/acre (±40% NDVI proxy)
30% ground cover	NDVI threshold	■ Estimated adequate cover zones based on NDVI threshold — field verification recommended

Satellite-verified cover crop status. Field verification recommended before termination.

Field Level Results

Metric	Value
NDVI Mean	0.251
NDVI Range	0.146 – 0.576
Slope Mean (%)	12.0%
C-Factor (exp. model)	0.185 (26% reduction vs. baseline)
Risk Index (C×LS)	0.592

Erosion Concern	**Moderate**

Metric	Value
Est. Cover Crop Erosion Reduction	23.4%
Baseline Soil Loss (no cover)	42.5 t/ac/yr
Est. Soil Saved (area-weighted)	10.0 t/ac/yr

Estimates based on RUSLE C-factor methodology. C-factor derived from piecewise exponential NDVI model. ±10 pt uncertainty on reduction percentage.

NDVI Source: Sentinel-2 via Google Earth Engine (NDVI: Apr 05, 2026 – Apr 20, 2026) | DEM: Iowa DNR 3m | Slope: computed in UTM meters (EPSG:26915)
Data provenance — DEM source: Iowa DNR 3m | R-factor source: R=146 (USDA AH703 erosivity, NOAA Midwest raster)
C-Factor methodology: piecewise exponential NDVI model — continuous C-factor differentiated by residue system. This report is advisory only and does not constitute an official NRCS determination.